# Mohammad VazirPanah

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Nationality: Iranian · DOB: 23 Apr 1994

## Summary

Embedded and cyber-physical systems researcher working on the design, formal modeling, and performance-efficient implementation of heterogeneous real-time systems, spanning formal CPS modeling and efficient execution on modern parallel and vector hardware (RISC-V). Particular interest in performance-portable execution of compute- and AI/ML-intensive workloads across heterogeneous architectures.

## Research Interests

Cyber-Physical & Embedded Systems, Real-Time Heterogeneous Computing, Formal Modeling & Verification, RISC-V / Vector Architectures, High-Performance Computing, Efficient AI & ML Acceleration on Heterogeneous Hardware

## Education

**M.Sc. in Computer Engineering — Computer Systems Architecture** *Sep 2019 – Oct 2022*
*Shahid Beheshti University, Tehran, Iran*

- GPA: 3.79 / 4.00
- Thesis: *A Framework for Modeling Self-Aware Cyber-Physical Systems with SystemC*
- Supervisor: Dr. Seyed-Hosein Attarzadeh-Niaki

**B.Sc. in Computer Engineering — Hardware Engineering** *Sep 2012 – May 2016*
*Birjand University, Birjand, Iran*

- GPA: 3.40 / 4.00
- Project: *Simulation and Programming the Alpha Rex Human-Robot Produced by Lego Company*

## Work Experience

**Research Fellow** *2023 – 2026*
*University of Salerno — Dept. of Computer Science, ISISLab, Salerno, Italy*

- Scientific Responsible: Prof. Biagio Cosenza
- Research bridging real-time embedded systems and high-performance heterogeneous computing across the RISC-V computing continuum, from embedded cores to HPC vector processors
- Benchmarked GCC, LLVM, and LLVM-EPI auto-vectorization for the RISC-V Vector Extension (RVV 0.7 & 1.0) across 151 TSVC loops and seven real-world applications on two RISC-V vector boards; showed LMUL tuning yields up to 3× speedup and characterized per-pattern compiler limitations. Published at IEEE PDP 2025 (2nd author)
- Contributed to LibreRT: developing and deploying portable real-time applications across heterogeneous embedded CPU/GPU platforms (SYCL, OpenMP, OpenCL)
- Built reproducible experiment pipelines (Git, Docker, CI, CMake) for RISC-V/SoC boards; documented setups for reproducibility

**Instructor** *2013 – 2015*
*Sepehr Educational Institutions, Kashmar, Iran*

- Delivered theoretical and practical instruction in C++ programming
- Taught ICDL (International Computer Driving License) skills
- Taught CorelDRAW and Adobe FreeHand, focusing on vector graphics

**Education Intern** *2014 – 2015*
*Abooreyhan Company, Kashmar, Iran*

- Hardware/software troubleshooting, diagnostics, and component replacement
- Assembled and configured PCs; software installation and system setup

## Academic Experience

### Teaching Assistantships

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|---|---|---|
| *Design & Analysis of Real-Time Embedded Systems* | SBU | 2021 |
| *System-on-Chip* | SBU | 2020, 2021 |
| *Advanced Computer Architecture* | SBU | 2021 |
| *Fundamentals of Embedded & Real-Time Systems* | SBU | 2020 |
| *Computer Architecture* | Birjand Univ. | 2016 |

### Laboratory Memberships

- ISISLab — HPC Group, University of Salerno, Italy
- SBU Cyber-Physical Systems Lab Research Team, Tehran, Iran

## Publications

**2026** **LibreRT: Portable Heterogeneous Real-Time Programming for the Embedded Computing Continuum**
*B. Cosenza, F. Terraneo, G. Agosta, L. Carpentieri, D. Cattaneo, **M. VazirPanah**, N. Nicolosi*
23rd ACM International Conference on Computing Frontiers (CF), Special Sessions · 2026

**2025** **A Performance Analysis of Autovectorization on RVV RISC-V Boards**
*L. Carpentieri, **M. VazirPanah**, B. Cosenza*
PDP — 33rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing, IEEE · 2025

**2023** **System-Level Modeling of Dynamic Applications with Scenario-Aware Dataflow Graphs**
*S.-H. Attarzadeh-Niaki, **M. VazirPanah***
Journal of Innovations in Computer Science and Engineering (JICSE), Shahid Beheshti University · 2023

**2022** **ROS-Based Co-Simulation for Formal Cyber-Physical Robotic System Design**
***M. VazirPanah**, S.-H. Attarzadeh-Niaki, A. Salimi-Badr*
27th International Computer Conference, Computer Society of Iran (CSICC), IEEE · 2022

## Selected Projects

**Self-Aware CPS Modeling Framework** *2022*
*M.Sc. Thesis Open-source project GitHub based on ForSyDe-SystemC Modeling Framework*

- Designed and implemented a self-aware cyber-physical system model using ForSyDe-SystemC and ROS-Gazebo co-simulation
- Developed an SADF-based model and contributed extensions to the ForSyDe-SystemC library

**LibreRT — Portable Heterogeneous Real-Time Programming for the Embedded Computing Continuum** *2023 – 2026*
*Researcher, Italian PRIN 2022-funded project*

- Developed and evaluated real-time applications for heterogeneous embedded CPU/GPU platforms
- Implemented solutions using SYCL, OpenMP, OpenCL, and RISC-V boards

**RISC-V Benchmark Suite** *2025*
*Open-source project GitHub*

- Designed and implemented an open-source benchmark suite for RISC-V and RVV architectures, covering diverse data-level-parallelism patterns
- Evaluated auto-vectorization performance across multiple GCC and LLVM compiler versions to isolate cost factors

**Railroad System Verification with NuSMV** *2020*
*Course Project, M.Sc. Design and Analysis of Real-Time Embedded Systems*

- Modeled and verified a railroad control system using NuSMV
- Checked safety and liveness properties through formal model checking

**FIR Filter and Tiny YOLO v1 on FPGA** *2019*

*Course Project, M.Sc. System-on-Chip, DE0-Nano / DE10 Intel FPGA Board*

- Designed and implemented a FIR filter IP core in Quartus with an Avalon-MM interface
- Deployed a real-time Tiny YOLO v1 object-detection (CNN) inference pipeline on FPGA under tight resource constraints; an edge-AI / ML inference workload on constrained hardware

## Technical Skills

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|---|---|
| **Programming** | C, C++, Python, Assembly |
| **Methods & Concepts** | Parallel & vector computing, performance portability, formal verification & model checking, real-time & embedded systems |
| **Modeling & Verification** | SystemC, NuSMV, UPPAAL |
| **Hardware & Embedded Systems** | RISC-V, FPGA, VHDL, Verilog, FreeRTOS |
| **Heterogeneous & HPC** | SYCL, OpenMP, OpenCL, CUDA |
| **Simulation & EDA** | Gem5, QEMU, OVPsim, Quartus, ModelSim, MATLAB-Simulink |
| **Robotics** | ROS, Gazebo, sensor integration |
| **Tools** | Git, Docker, Linux, CMake, LaTeX |
| **Certifications** | TinyML — Harvard University, edX |

## Honours and Awards

**National R&D Research Grant Winner** *2020*
*R&D Center, Mobile Telecommunication Company of Iran (MCI)*

- Awarded among competing Master's and PhD students in a national grant competition

**Top Computer Engineering Student** *2016*
*Birjand University*

- Ranked in the top 15% of B.Sc. Computer Engineering graduates

## Conference Participation

**33rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP)** *Mar 2025*
*Co-author and attendee — Turin, Italy*

- Co-author of the paper: *A Performance Analysis of Autovectorization on RVV RISC-V Boards*

**27th International Computer Conference, Computer Society of Iran (CSICC)** *Feb 2022*
*Presenter — Sharif University of Technology, Tehran, Iran*

- Presented the paper: *ROS-Based Co-Simulation for Formal Cyber-Physical Robotic System Design*

## Language Skills

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|---|---|
| **Persian** | Native |
| **English** | Proficient — (Listening, Reading, Writing, Speaking) |
| **Italian** | Basic — (Listening, Reading, Writing, Speaking) |

## References


**Prof. Biagio Cosenza**
*Research Supervisor*
University of Salerno, Italy
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**Prof. Vittorio Scarano**
*Head of ISISLab*
University of Salerno, Italy
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**Dr. S.-H. Attarzadeh-Niaki**
*M.Sc. Supervisor*
Shahid Beheshti University, Iran
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